

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-4

DURATION: 30 MINUTES

SUMMER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. State the fundamental theorems of calculus. Applying these theorems, explain how Euler's method is used to approximate the value of a definite integral. 3 + 2
(CO4)
(PO1)

Solution:

The fundamental theorems of calculus are –

- i. Let f be a continuous real-valued function defined on a closed interval $[a, b]$. Let F be the function defined, for all x in $[a, b]$ by

$$F(x) = \int_a^x f(t) dt$$

Then F is uniformly continuous on $[a, b]$ and differentiable on the open interval (a, b) , and $F'(x) = f(x)$ for all x in (a, b) so F is an antiderivative of f .

- ii. If f is Riemann integrable on $[a, b]$ then

$$\int_a^b f(x) dx = F(b) - F(a)$$

The idea is to rewrite the integral as the solution of an ODE,

$$\frac{dy}{dx} = f(x), y(a) = 0, \text{ using the 1st Fundamental theorem}$$

where then $y(b)$ gives us the value of the integral $\int_a^b f(x) dx$, according to the 2nd Fundamental theorem. We can use Euler's method to approximate the value of the dependent variable y at the discrete point $x = b$.

Rubric:

- 1.5 points each correct fundamental theorem.
- 2 points for the correct explanation.

2. Compare and contrast the following numerical algorithms for solving 1st order ODEs – Euler’s method, Runge-Kutta 2nd order method, and Runge-Kutta 4th order method.

2
(CO3)
(PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Euler	Runge-Kutta 2nd Order	Runge-Kutta 4th Order
Formula	$y_{i+1} = y_i + f(x_i, y_i)h$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2)h, k_1 = f(x_i, y_i), k_2 = f(x_i + ph, y_i + qk_1h)$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2 + a_3k_3 + a_4k_4)h$
Terms of the Taylor series	First 2 terms.	First 3 terms.	First 5 terms.
Order of the Taylor polynomial	1st order – which is why it’s also called the Runge-Kutta 1st order method.	2nd order – which is why it’s called the Runge-Kutta 2nd order method.	4th order – which is why it’s also called the Runge-Kutta 4th order method.
Local Truncation Error	$O(h^2)$	$O(h^3)$	$O(h^5)$
Global Truncation Error	$O(h)$	$O(h^2)$	$O(h^4)$
Variant Approaches	None.	Has 3 different versions – Heun’s method, Midpoint method, and Ralston’s method	Has 2 different versions – Runge’s approach and Kutta’s approach.

Rubric:

- 1 point for each valid comparison.

3. Observe the lightly shaded solid object portrayed in Figure 1.

Its base is enclosed by a curve $y = \sqrt{x} + 1$, the x -axis, the y -axis, and the straight line $x = 1$. When this solid object is cut by a plane perpendicular to the x -axis and the xy -plane, the resulting cross-section is a square (as indicated by the darkly shaded cross-sectional area).

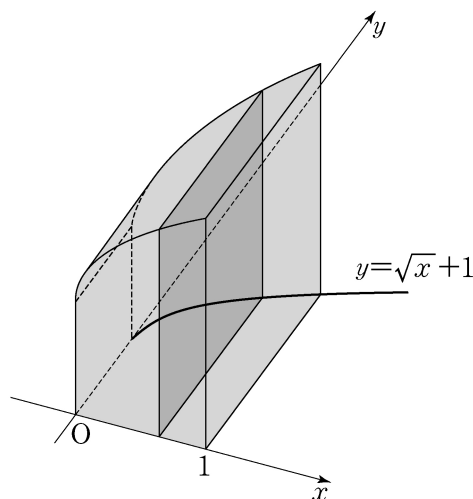


Figure 1: A solid geometric object.

- a) Formulate an integrand function $f(x)$ whose integral value within the limits $[0, 1]$ will yield the volume of the solid object.

1.5
(CO1)
(PO1)

Solution:

There are 2 key observations here—

- The cross-sections of the object are squares.
- The sides of the squares are gradually increasing according to the increasing nature of the $y = \sqrt{x} + 1$ curve.

We know, that the area of a square having a side of length s is s^2 .

Here, for a particular value of x , the side length of an infinitesimally thin square cross-sectional slice is $\sqrt{x} + 1$. So, the area of that square slice will be $(\sqrt{x} + 1)^2$. All of these infinitesimally thin square slices within the limits $[0, 1]$ together constitute the entire volume of the solid object, *ergo* the integrand is,

$$f(x) = (\sqrt{x} + 1)^2$$

Rubric:

- 1.5 points for a correct integrand function.

- b) Approximate the volume of the solid object using Mixed Simpson's rule *i.e.* Simpson's 1/3 rule (with $n_1 = 6$) and Simpson's 3/8 rule (with $n_2 = 6$).

4.5
(CO2)
(PO1)

Solution:

Here, $h = \frac{b-a}{n} = \frac{b-a}{n_1+n_2} = \frac{1-0}{12} \approx 0.08334$. Let's compute the functional values at the right-hand side of each of these $n = 12$ segments.

$x_0 = a = 0;$	$f(x_0) = 1$
$x_1 = a + h = 0 + 0.08334;$	$f(x_1) = 1.6607$
$x_2 = x_1 + h = 0.1666;$	$f(x_2) = 1.9832$
$x_3 = x_2 + h = 0.25;$	$f(x_3) = 2.25$
$x_4 = x_3 + h = 0.3333;$	$f(x_4) = 2.488$
$x_5 = x_4 + h = 0.4166;$	$f(x_5) = 2.7076$
$x_6 = x_5 + h = 0.5;$	$f(x_6) = 2.9142$
$x_7 = x_6 + h = 0.5833;$	$f(x_7) = 3.1108$
$x_8 = x_7 + h = 0.6666;$	$f(x_8) = 3.2996$
$x_9 = x_8 + h = 0.75;$	$f(x_9) = 3.482$
$x_{10} = x_9 + h = 0.8333;$	$f(x_{10}) = 3.659$
$x_{11} = x_{10} + h = 0.9166;$	$f(x_{11}) = 3.8315$
$x_{12} = x_{11} + h = 1;$	$f(x_{12}) = 4$

The multiple-segment equation for Simpson's 1/3 rule is

$$I_1 = \int_a^b f(x)dx \approx \frac{b-a}{3n_1} \left[f(x_0) + 4 \sum_{\substack{i=1, \\ i=\text{odd}}}^{n_1-1} f(x_i) + 2 \sum_{\substack{i=2, \\ i=\text{even}}}^{n_1-2} f(x_i) + f(x_{n_1}) \right]$$

where, $a = 0$ and $b = x_6 = 0.5$. Plugging in the respective values, we get,

$$I_1 \approx \frac{0.5}{3 \times 6} [1 + 4 \times (1.6607 + 2.25 + 2.7076) + 2 \times (1.9832 + 2.488) + 2.9142] \\ = 1.0925$$

The multiple-segment equation for Simpson's 3/8 rule is

$$I_2 = \int_a^b f(x)dx \approx \frac{3 \times (b-a)}{8n_2} \left[f(x_6) + 3 \sum_{\substack{i=1, \\ i\%3 \neq 0}}^{n_1+n_2-1} f(x_i) + 2 \sum_{\substack{i=3, \\ i\%3=0}}^{n_1+n_2-3} f(x_i) + f(x_{n_1+n_2}) \right]$$

where, $a = x_6 = 0.5$ and $b = x_{12} = 1$. Plugging in the respective values, we get,

$$I_2 \approx \frac{3 \times 0.5}{8 \times 6} [2.9142 + 3 \times (3.1108 + 3.2996 + 3.659 + 3.8315) + 2 \times (3.482) + 4] \\ = 1.7369$$

So the volume of the solid object is $I = \int_0^1 (\sqrt{x} + 1)^2 dx \approx I_1 + I_2 = 1.0925 + 1.7369 = 2.8294$.

Rubric:

- 2 points for a correct I_1 value.
- 2 points for a correct I_2 value.
- 0.5 point for an approximately correct answer.

4. An internet meme is defined as a humorous or culturally relevant image, video, or text that spreads rapidly online, often evolving as it is shared and adapted by users. Draw a rough sketch of a math meme related to this course. 2
(CO69)
(PO420)

Warning: Cringe and lame memes will result in negative marking.

Solution:



Figure 2: Real.

Rubric:

- 2 points for a dank meme.